Rigid-body Dynamics

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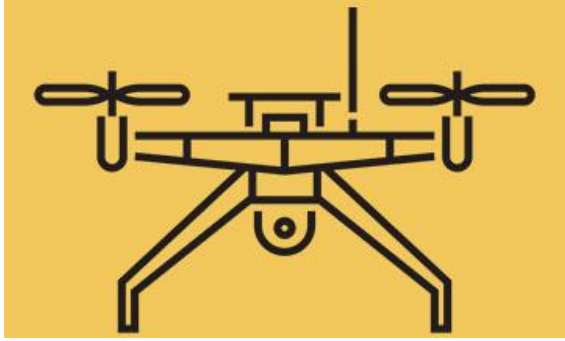
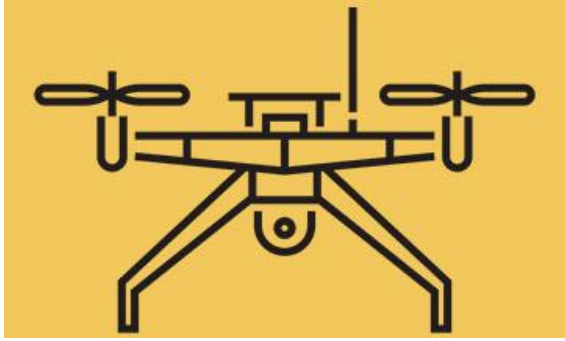
SDU

Today's lecture

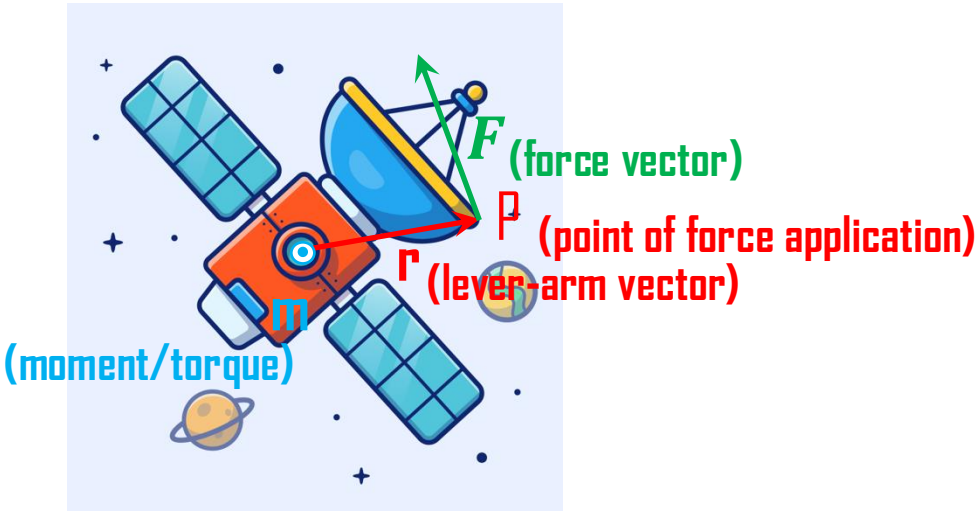


- Forces and moments
- More kinematics: rotations without Euler angles
- Transport theorem and motion in body-fixed coordinates
- The inertia tensor and Euler's rotation equation
- Implementation aspects
- Hybrid model for multi-rotors

Fundamentals of classical mechanics



Forces and moments

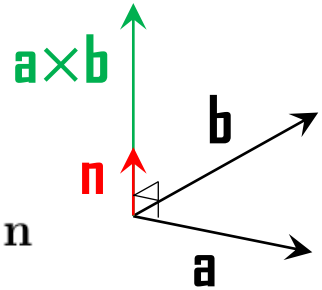


$$\boxed{\mathbf{m} = \mathbf{r} \times \mathbf{F}}$$

Uniform circular motion around fixed axis

$$\boxed{\mathbf{v}_\perp = \frac{d\mathbf{r}}{dt} = \boldsymbol{\omega} \times \mathbf{r}}$$

Recall vectorial/cross product:



$$\mathbf{a} \times \mathbf{b} = ||\mathbf{a}|| \cdot ||\mathbf{b}|| \sin\theta \cdot \mathbf{n}$$

$$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$$

$$\mathbf{a} \times \mathbf{a} = 0$$

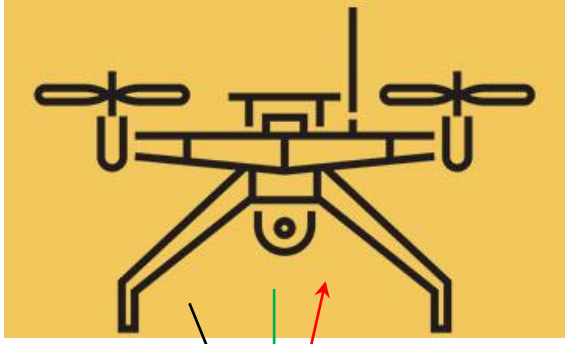
$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$$

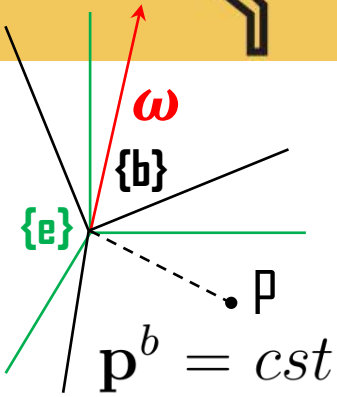
Skew-symmetric matrix

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = [\mathbf{a}]_\times \mathbf{b}$$

(and use the circular motion concept seen in the previous slide)

Parenthesis: back to kinematics (1/2)





$$\mathbf{p}^b = cst$$

- Consider a point P, fixed in the body-fixed frame
- Frame {b} is rotating with angular velocity $\boldsymbol{\omega}$ with respect to frame {e}.

Transforming between {b} and {e} frames gives

$$\mathbf{p}^e = \mathbf{R}_b^e \mathbf{p}^b$$

and $$\boldsymbol{\omega}_{b/e}^e = \mathbf{R}_b^e \boldsymbol{\omega}_{b/e}^b$$

Velocity of $\mathbf{p}^e$:

$$\frac{d}{dt}\mathbf{p}^e = \frac{d}{dt}\left(\mathbf{R}_b^e\,\mathbf{p}^b\right)$$

$$= \frac{d}{dt}\left(\mathbf{R}_b^e\right)\mathbf{p}^b + \mathbf{R}_b^e \underbrace{\cancel{\frac{d}{dt}\left(\mathbf{p}^b\right)}}_{=0}$$

{b} is a rotating frame: $\left(\frac{d\mathbf{r}}{dt} = \boldsymbol{\omega} \times \mathbf{r}\right)$

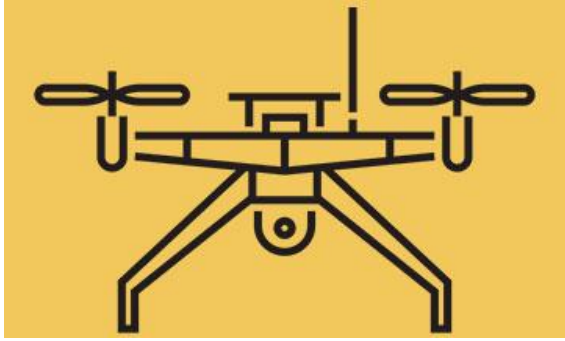
$$\frac{d}{dt}\mathbf{p}^e = \boldsymbol{\omega}_{b/e}^e \times \mathbf{p}^e$$

$$= \left[\boldsymbol{\omega}_{b/e}^e\right]_\times \mathbf{p}^e$$

$$= \left[\boldsymbol{\omega}_{b/e}^e\right]_\times \mathbf{R}_b^e\,\mathbf{p}^b$$

$$\frac{d}{dt}\left(\mathbf{R}_b^e\right)\mathbf{p}^b = \left[\boldsymbol{\omega}_{b/e}^e\right]_\times \mathbf{R}_b^e\,\mathbf{p}^b$$

Parenthesis: back to kinematics (2/2)



$$\frac{d}{dt}\left(\mathbf{R}_b^e\right)\mathbf{p}^b = \left[\boldsymbol{\omega}_{b/e}^e\right]_\times \mathbf{R}_b^e\,\mathbf{p}^b$$

is valid for all $\mathbf{p}^b$

Hence $$\boxed{\frac{d}{dt}\left(\mathbf{R}_b^e\right) = \left[\boldsymbol{\omega}_{b/e}^e\right]_\times \mathbf{R}_b^e}$$

BUT $\boldsymbol{\omega}_{b/e}$ is usually expressed/measured in the {b} frame.

$$\Rightarrow \frac{d}{dt}\left(\mathbf{R}_b^e\right) = \left[\mathbf{R}_b^e\,\boldsymbol{\omega}_{b/e}^b\right]_\times \mathbf{R}_b^e$$
$$= \mathbf{R}_b^e\left[\boldsymbol{\omega}_{b/e}^b\right]_\times \left(\mathbf{R}_b^e\right)^T \mathbf{R}_b^e$$

because $\left[\mathbf{R}\mathbf{a}\right]_\times = \mathbf{R}\left[\mathbf{a}\right]_\times \mathbf{R}^T$

so that we finally have $$\boxed{\dot{\mathbf{R}}_b^e = \mathbf{R}_b^e\left[\boldsymbol{\omega}_{b/e}^b\right]_\times}$$ with $\dot{\mathbf{R}}_b^e := \frac{d}{dt}\left(\mathbf{R}_b^e\right)$

Fundamental kinematic equation (matrix ODE)

Remarks: - Contrary to $\dot{\Theta} = \mathbf{T}_\Theta(\Theta)\boldsymbol{\omega}_{b/e}^b$, it does not have any singularity!

- the above expression does not use Euler angles at all...

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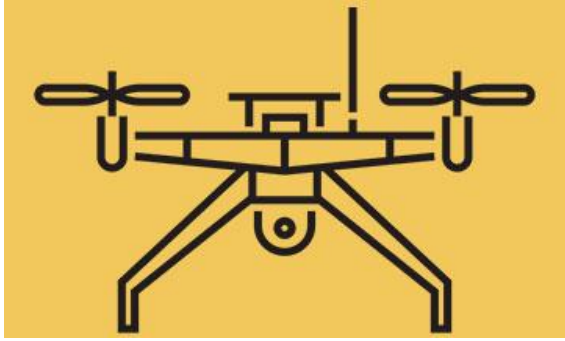
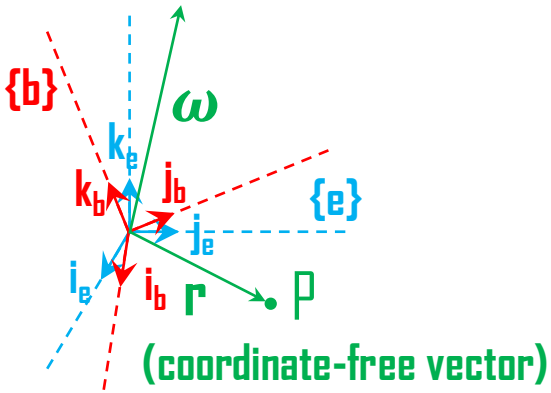
Derivative of a vector: earth- and body- frames (1/2)





$\mathbf{i}_e$, $\mathbf{j}_e$ and $\mathbf{k}_e$
$\mathbf{i}_b$, $\mathbf{j}_b$ and $\mathbf{k}_b$ unit vectors

Vector r will have coordinates $\mathbf{r} = x^e\mathbf{i}_e + y^e\mathbf{j}_e + z^e\mathbf{k}_e$ in the earth frame

$\mathbf{r} = x^b\mathbf{i}_b + y^b\mathbf{j}_b + z^b\mathbf{k}_b$ in the body frame

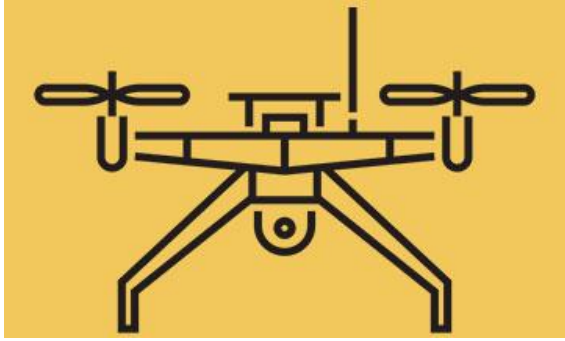
Hence (since the earth-fixed frame is not moving)

$$\frac{d\mathbf{r}}{dt} = \frac{dx^e}{dt}\mathbf{i}_e + \frac{dy^e}{dt}\mathbf{j}_e + \frac{dz^e}{dt}\mathbf{k}_e \quad (1)$$

while in the body/rotational frame

$$\frac{d\mathbf{r}}{dt} = \frac{dx^b}{dt}\mathbf{i}_b + \frac{dy^b}{dt}\mathbf{j}_b + \frac{dz^b}{dt}\mathbf{k}_b + x^b\frac{d\mathbf{i}_b}{dt} + y^b\frac{d\mathbf{j}_b}{dt} + z^b\frac{d\mathbf{k}_b}{dt} \quad (2)$$

Derivative of a vector: earth- and body- frames (2/2)



Putting (1) and (2) together, we have

$$\underbrace{\frac{dx^e}{dt}\mathbf{i}_e + \frac{dy^e}{dt}\mathbf{j}_e + \frac{dz^e}{dt}\mathbf{k}_e}_{\frac{{}^e d}{dt}\mathbf{r} :=} = \underbrace{\frac{dx^b}{dt}\mathbf{i}_b + \frac{dy^b}{dt}\mathbf{j}_b + \frac{dz^b}{dt}\mathbf{k}_b}_{\frac{{}^b d}{dt}\mathbf{r} :=} + \underbrace{x^b\frac{d\mathbf{i}_b}{dt} + y^b\frac{d\mathbf{j}_b}{dt} + z^b\frac{d\mathbf{k}_b}{dt}}_{= x^b(\boldsymbol{\omega}_{b/e}\times\mathbf{i}_b) + y^b(\boldsymbol{\omega}_{b/e}\times\mathbf{j}_b) + z^b(\boldsymbol{\omega}_{b/e}\times\mathbf{k}_b)}$$

because $\frac{d\mathbf{r}}{dt} = \boldsymbol{\omega}\times\mathbf{r}$

$$= \boldsymbol{\omega}_{b/e}\times(x^b\mathbf{i}_b) + \boldsymbol{\omega}_{b/e}\times(y^b\mathbf{j}_b) + \boldsymbol{\omega}_{b/e}\times(z^b\mathbf{k}_b)$$

$$= \boldsymbol{\omega}_{b/e}\times\mathbf{r}$$

So that we finally have

$$\boxed{\frac{{}^e d}{dt}\mathbf{r} = \frac{{}^b d}{dt}\mathbf{r} + \boldsymbol{\omega}_{b/e}\times\mathbf{r}}$$

Transport theorem

Velocity of point P as seen from the body-fixed frame

Velocity of point P as seen from the earth-fixed frame

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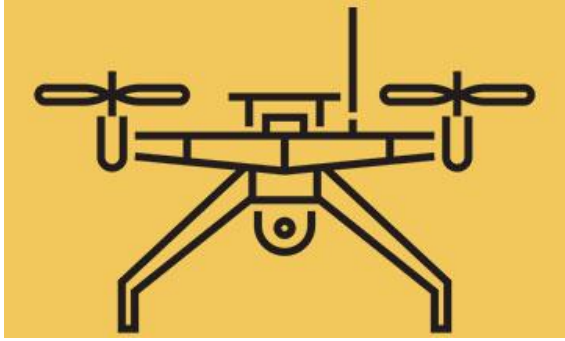
Linear motion of a rigid body



(in an inertial frame,) we have conservation of the linear momentum of a rigid body, so that Newton's law can be rewritten as:

$$\frac{d}{dt}\boldsymbol{\mu} = \mathbf{F} \quad \text{with} \quad \boldsymbol{\mu} := m\mathbf{v}_{b/e}$$

Using the <u>transport theorem</u>, we have

$$\begin{aligned}\frac{{}^{e}d}{dt}(m\mathbf{v}_{b/e}) &= \frac{{}^{b}d}{dt}(m\mathbf{v}_{b/e}) + \boldsymbol{\omega}_{b/e} \times (m\mathbf{v}_{b/e}) \\ &= m\dot{\mathbf{v}}_{b/e} + \boldsymbol{\omega}_{b/e} \times (m\mathbf{v}_{b/e})\end{aligned}$$

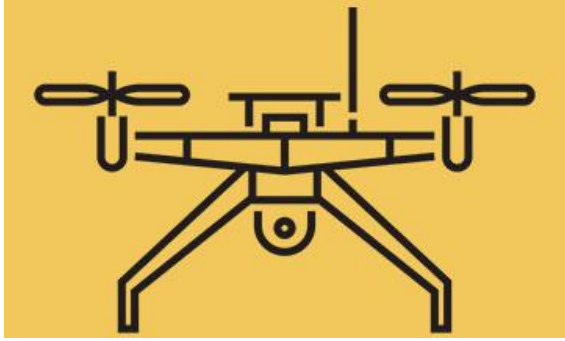
$$\text{with} \quad \dot{\mathbf{v}} := \frac{{}^{b}d}{dt}\mathbf{v}$$

Expressed in the body-fixed coordinates, we then have

$$\boxed{m\dot{\mathbf{v}}^{b}_{b/e} + \boldsymbol{\omega}^{b}_{b/e} \times (m\mathbf{v}^{b}_{b/e}) = \mathbf{F}^{b}_{c}}$$

Force vector applied to body center of gravity, expressed in the frame {b}

Angular momentum



For rotational motion, the momentum equation reads

$$\frac{d}{dt}\mathbf{h} = \mathbf{m}$$ in the inertial/earth-fixed frame

Angular momentum

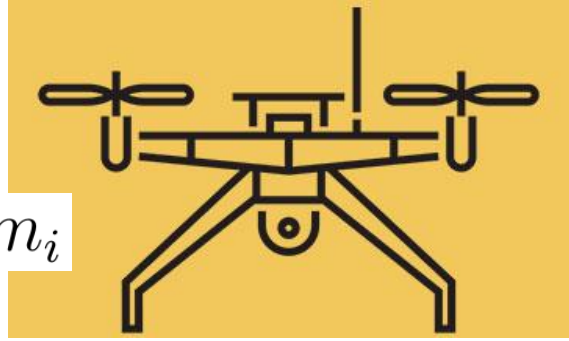
Using the <u>transport theorem</u> again, we have

$$\frac{{}^{e}d}{dt}\mathbf{h} = \frac{{}^{b}d}{dt}\mathbf{h} + \boldsymbol{\omega}_{b/e} \times \mathbf{h}$$

So that we get

$$\boxed{\frac{{}^{b}d}{dt}\mathbf{h} + \boldsymbol{\omega}_{b/e} \times \mathbf{h} = \mathbf{m}}$$

The inertia matrix/tensor (1/2)



Consider a body modelled by a collection of n mass points m_i whose coordinates $\mathbf{r}_i^b$ are expressed in the body-fixed coordinates.

(O_b is assumed to be the center of gravity of the body)

Then, the angular momentum is given by

$$\mathbf{h}^b = \sum_{i=1}^{n} \mathbf{r}_i^b \times \boldsymbol{\mu}_i^b$$

because, for a single particle:

$$\mathbf{h}^b = \mathbf{r}^b \times \boldsymbol{\mu}^b$$

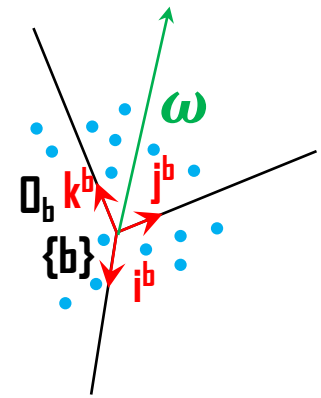
(relation similar to between moment and force, see s. 3)

$$= \sum_{i=1}^{n} \mathbf{r}_i^b \times (m_i \mathbf{v}_i^b)$$

$$= \sum_{i=1}^{n} \mathbf{r}_i^b \times m_i(\boldsymbol{\omega}_{b/e}^b \times \mathbf{r}_i^b)$$

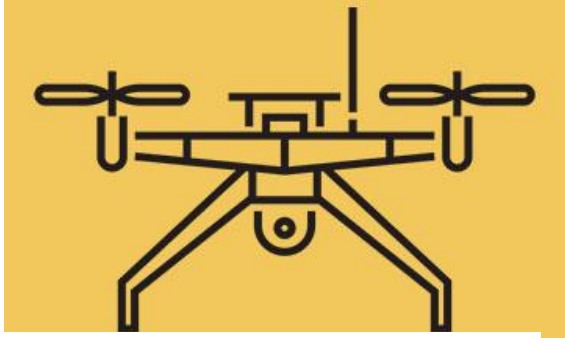
$$= \sum_{i=1}^{n} m_i \left[\mathbf{r}_i^b \times (\boldsymbol{\omega}_{b/e}^b \times \mathbf{r}_i^b)\right]$$

$$= \sum_{i=1}^{n} m_i \left[(\mathbf{r}_i^b \cdot \mathbf{r}_i^b)\boldsymbol{\omega}_{b/e}^b - (\mathbf{r}_i^b \cdot \boldsymbol{\omega}_{b/e}^b)\mathbf{r}_i^b\right]$$



because $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

The inertia matrix/tensor (2/2)



Let $\mathbf{r}_i^b = \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix}$ and $\boldsymbol{\omega}_{b/e}^b = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$

Then $$(\mathbf{r}_i^b \cdot \mathbf{r}_i^b)\boldsymbol{\omega}_{b/e}^b - (\mathbf{r}_i^b \cdot \boldsymbol{\omega}_{b/e}^b)\mathbf{r}_i^b = (x_i^2 + y_i^2 + z_i^2)\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} - (\omega_x x_i + \omega_y y_i + \omega_z z_i)\begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix}$$

$$= \begin{bmatrix} (x_i^2 + y_i^2 + z_i^2)\omega_x - (\omega_x x_i + \omega_y y_i + \omega_z z_i)x_i \\ (x_i^2 + y_i^2 + z_i^2)\omega_y - (\omega_x x_i + \omega_y y_i + \omega_z z_i)y_i \\ (x_i^2 + y_i^2 + z_i^2)\omega_z - (\omega_x x_i + \omega_y y_i + \omega_z z_i)z_i \end{bmatrix}$$

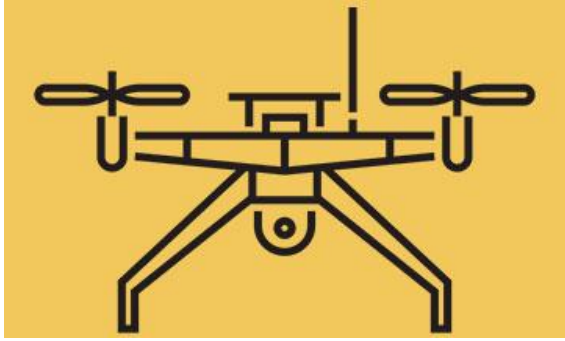
Hence $$\mathbf{h}^b = \sum_{i=1}^{n} m_i \begin{bmatrix} y_i^2 + z_i^2 & -x_i y_i & -x_i z_i \\ -x_i y_i & x_i^2 + z_i^2 & -y_i z_i \\ -x_i z_i & -y_i z_i & x_i^2 + y_i^2 \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

so that we have $$\mathbf{h}^b = \underbrace{\begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix}}_{\mathbf{I}_C} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

$\mathbf{I}_C$ inertia matrix/tensor (constant in the body frame)

so that $$\boxed{\mathbf{h}^b = \mathbf{I}_C\, \boldsymbol{\omega}_{b/e}^b}$$

Euler's rotation equations



The angular momentum (slide 9) expressed in the body coordinates is given by

$$\frac{{}^b d}{dt}\mathbf{h}^b + \boldsymbol{\omega}^b_{b/e} \times \mathbf{h}^b = \mathbf{m}^b_C$$

And using inertia tensor expression (slide 11), we get

$$\frac{{}^b d}{dt}(\mathbf{I}_C \boldsymbol{\omega}^b_{b/e}) + \boldsymbol{\omega}^b_{b/e} \times (\mathbf{I}_C \boldsymbol{\omega}^b_{b/e}) = \mathbf{m}^b_C$$

and $\mathbf{I}_C \frac{{}^b d}{dt}\boldsymbol{\omega}^b_{b/e} + \boldsymbol{\omega}^b_{b/e} \times (\mathbf{I}_C \boldsymbol{\omega}^b_{b/e}) = \mathbf{m}^b_C$ since inertia is constant in the body frame

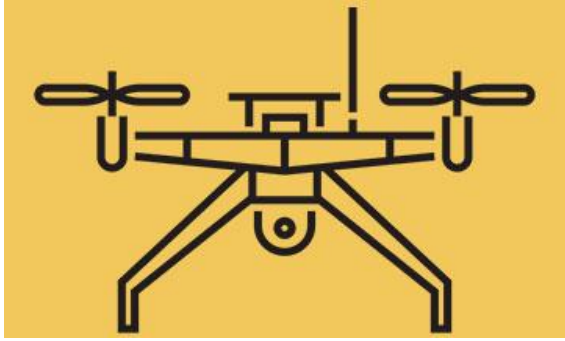
Hence, we finally get

$$\boxed{\mathbf{I}_C \dot{\boldsymbol{\omega}}^b_{b/e} + \boldsymbol{\omega}^b_{b/e} \times (\mathbf{I}_C \boldsymbol{\omega}^b_{b/e}) = \mathbf{m}^b_C}$$

Euler's rotation equations

with $\dot{\boldsymbol{\omega}}^b_{b/e} := \frac{{}^b d}{dt}\boldsymbol{\omega}^b_{b/e}$

Towards implementation (1/2)



We have

$$\begin{cases} m\dot{\mathbf{v}}^b_{b/e} + \boldsymbol{\omega}^b_{b/e} \times (m\mathbf{v}^b_{b/e}) = \mathbf{F}^b_c \\ \mathbf{I}_C\dot{\boldsymbol{\omega}}^b_{b/e} + \boldsymbol{\omega}^b_{b/e} \times (\mathbf{I}_C\boldsymbol{\omega}^b_{b/e}) = \mathbf{m}^b_C \end{cases}$$

$$\mathbf{I}_C\dot{\boldsymbol{\omega}}^b_{b/e} - (\mathbf{I}_C\boldsymbol{\omega}^b_{b/e}) \times \boldsymbol{\omega}^b_{b/e} = \mathbf{m}^b_C$$

because $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$

Using the matrix form of the cross product

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = [\mathbf{a}]_\times \mathbf{b}$$

and putting things in vectorial form, the above can be rewritten as

$$\boxed{\begin{bmatrix} m\boldsymbol{I}_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & \mathbf{I}_C \end{bmatrix} \begin{bmatrix} \dot{\mathbf{v}}^b_{b/e} \\ \dot{\boldsymbol{\omega}}^b_{b/e} \end{bmatrix} + \begin{bmatrix} m\left[\boldsymbol{\omega}^b_{b/e}\right]_\times & 0_{3\times3} \\ 0_{3\times3} & -\left[\mathbf{I}_C\boldsymbol{\omega}^b_{b/e}\right]_\times \end{bmatrix} \begin{bmatrix} \mathbf{v}^b_{b/e} \\ \boldsymbol{\omega}^b_{b/e} \end{bmatrix} = \begin{bmatrix} \mathbf{F}^b_C \\ \mathbf{m}^b_C \end{bmatrix}}$$

Inertia matrix (invertible)

Coriolis matrix

Generalized velocities

Forces and torques vector

$$\boxed{\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}}$$

Rigid-body dynamics

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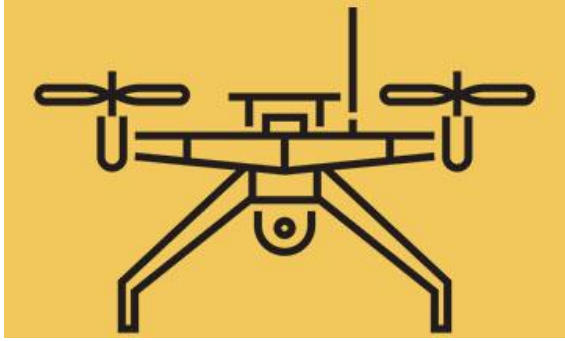
Towards implementation (2/2)



To bring $\boldsymbol{\nu} = \begin{bmatrix} \mathbf{v}^b_{b/e} \\ \boldsymbol{\omega}^b_{b/e} \end{bmatrix}$ back to an inertial frame, we can use

$$\begin{cases} \dot{\mathbf{p}}^e_{b/e} = \mathbf{R}^e_b(\boldsymbol{\Theta})\mathbf{v}^b_{b/e} \\ \dot{\boldsymbol{\Theta}} = \mathbf{T}_{\boldsymbol{\Theta}}(\boldsymbol{\Theta})\boldsymbol{\omega}^b_{b/e} \end{cases}$$

or

$$\begin{cases} \dot{\mathbf{p}}^e_{b/e} = \mathbf{R}^e_b\,\mathbf{v}^b_{b/e} \\ \dot{\mathbf{R}}^e_b = \mathbf{R}^e_b\left[\boldsymbol{\omega}^b_{b/e}\right]_\times \end{cases}$$

Not singular, does not depend on choice of Euler angles, but can drift a bit over time

$$\boxed{\begin{bmatrix} \dot{\mathbf{p}}^e_{b/e} \\ \dot{\boldsymbol{\Theta}} \end{bmatrix} = \begin{bmatrix} \mathbf{R}^e_b(\boldsymbol{\Theta}) & 0_{3\times3} \\ 0_{3\times3} & \mathbf{T}_{\boldsymbol{\Theta}}(\boldsymbol{\Theta}) \end{bmatrix} \begin{bmatrix} \mathbf{v}^b_{b/e} \\ \boldsymbol{\omega}^b_{b/e} \end{bmatrix}}$$

i.e.

$$\boxed{\dot{\boldsymbol{\eta}} = \mathbf{J}_{\boldsymbol{\Theta}}(\boldsymbol{\eta})\boldsymbol{\nu}}$$

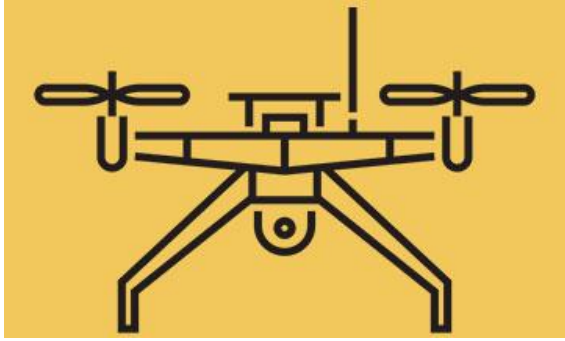
Kinematics

Together with the rigid-body dynamics to get

$$\boxed{\begin{cases} \dot{\boldsymbol{\eta}} = \mathbf{J}_{\boldsymbol{\Theta}}(\boldsymbol{\eta})\boldsymbol{\nu} \\ \mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau} \end{cases}}$$

Valid except for $\boldsymbol{\eta}$ values where $\mathbf{J}_{\boldsymbol{\Theta}}(\boldsymbol{\eta})$ is singular

Hybrid representation used for multi-rotors



Instead of expressing everything in the body-fixed frame, many studies/algorithms for multi-rotors use the hybrid representation:

$$\boxed{\begin{cases} m\dot{\mathbf{v}}^e_{b/e} = \mathbf{F}^e_c \\ \mathbf{I}_C\dot{\boldsymbol{\omega}}^b_{b/e} + \boldsymbol{\omega}^b_{b/e} \times (\mathbf{I}_C\boldsymbol{\omega}^b_{b/e}) = \mathbf{m}^b_C \end{cases}}$$

i.e. linear dynamics in <u>earth-fixed</u> frame
angular dynamics in <u>body-fixed</u> frame

with the transformation between positions and velocities

$$\begin{cases} \dot{\mathbf{p}}^e_{b/e} = \mathbf{v}^e_{b/e} \\ \dot{\boldsymbol{\Theta}} = \mathbf{T}_{\boldsymbol{\Theta}}(\boldsymbol{\Theta})\boldsymbol{\omega}^b_{b/e} \end{cases}$$

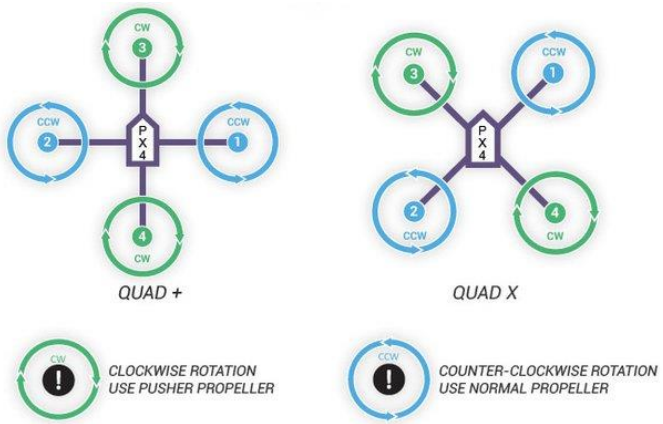
only derivative, not rotation

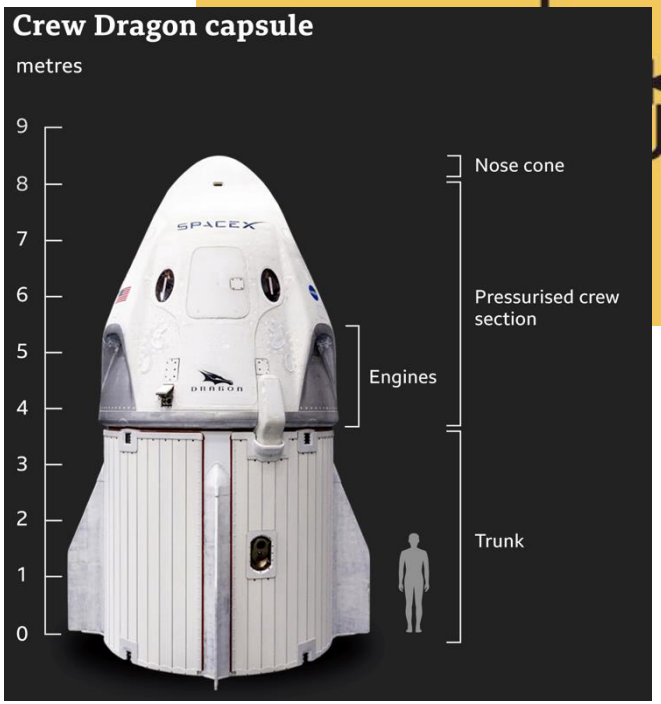
$$\boxed{\begin{cases} m\ddot{\mathbf{p}}^e_{b/e} = \mathbf{F}^e_c \\ \mathbf{I}_C\dot{\boldsymbol{\omega}}^b_{b/e} + \boldsymbol{\omega}^b_{b/e} \times (\mathbf{I}_C\boldsymbol{\omega}^b_{b/e}) = \mathbf{m}^b_C \end{cases}}$$

rewrite model further

Let us take a break

Today's exercise (1/3)





- Q1) Assuming a FRD frame for each type of drone frame (quad + or quad X), calculate, for each drone type, its inertia matrix with the distance between each motor and the CoG assumed to be 1m, while each motor is assumed to weigh 1kg
- Q2) Assuming that you can approximate the Crew Dragon capsule (see above picture) by a cylinder and using information you can get on the internet (ie weight, dimensions of vehicle, inertia tensor of cylinder), give an estimate of the inertia tensor of this spacecraft.

Today's exercise (2/3)

- Q3) From Itslearning, download and extract the zip file Spaceship_files. In Simulink, open the file spaceship_temp.slx and run it to make sure that it works.
- Q4) Using a body-fixed representation, implement the rigid-body dynamics of your spacecraft using the inertia tensor you previously found. Add the kinematics based on the Euler angles for the change of coordinates.
- Q5) Starting from a specific initial condition, apply pre-defined forces and moments with input blocks so that you start controlling your spacecraft.

Today's exercise (3/3)

- Q6) Test different initial conditions of the system so that it goes straight, make a rotation around a specific axis, a combination of both, etc
- Q7) Start from initial Euler angles [0,90,0] (in deg.). What happens when applying a torque to change ω_z ?
- Q8) Replace the Euler kinematics by the Rotation one (s. 14) and rerun the previous experiment. What happens?
- Q9) Replace the above blocks by real-time inputs such as gamepad commands or sliders.